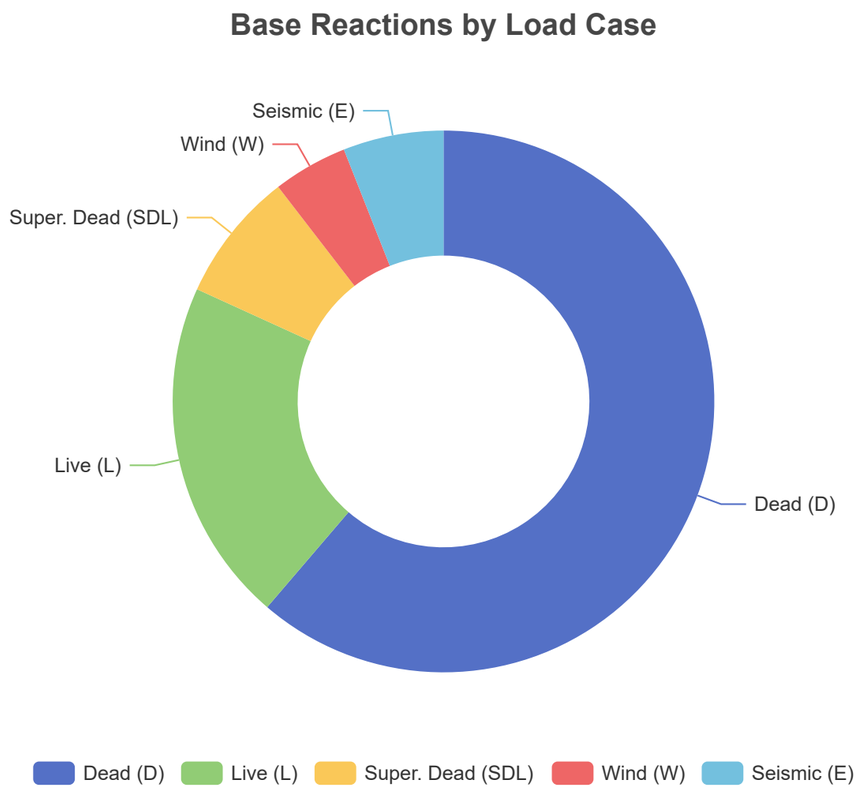
Pacific Tower

Foundation Design Report

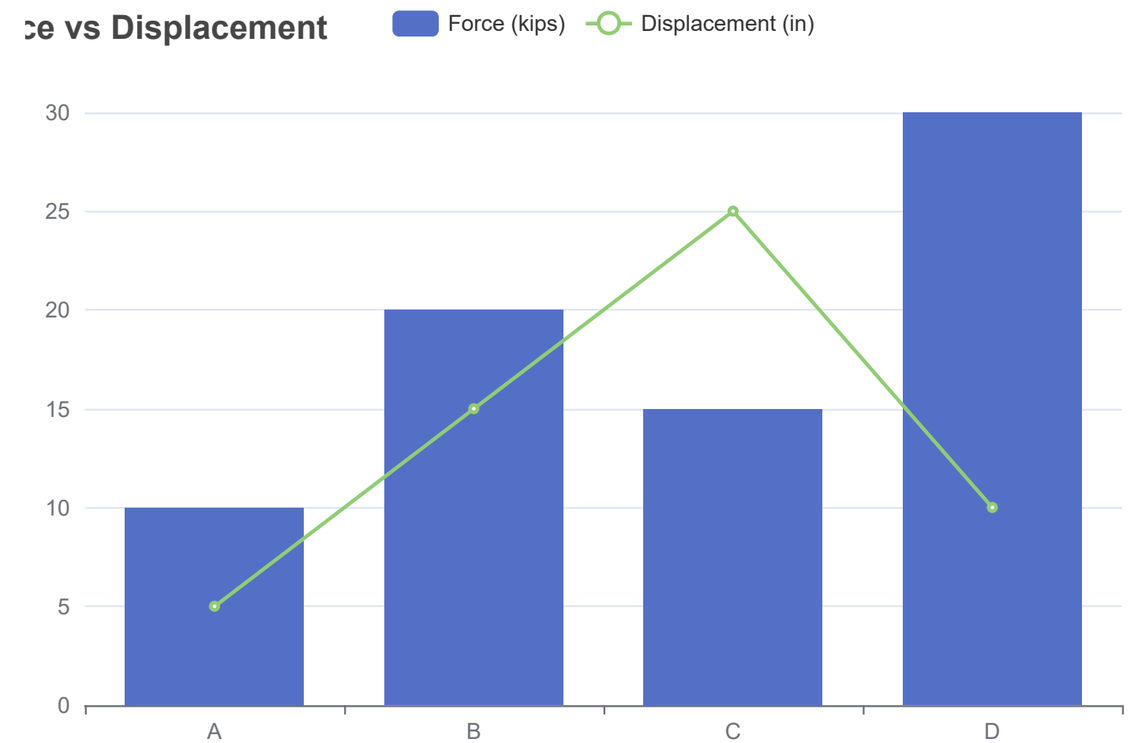
Foundation Design — Gravity & Lateral

Thornton Tomasetti	PROJECT Pacific Tower PROJECT NO. 2025-0042	DRAWING TITLE Foundation Design — Gravity & Lateral	REFERENCE S-FOUND-01 REVISION 0	DRAWN BY BT CHECKED BY RS DATE 2026-04-06	SCALE NTS SHEET SK-01

STRUCTURAL ANALYSIS — LOAD DISTRIBUTION



Base Reactions by Load Case



Force vs Displacement

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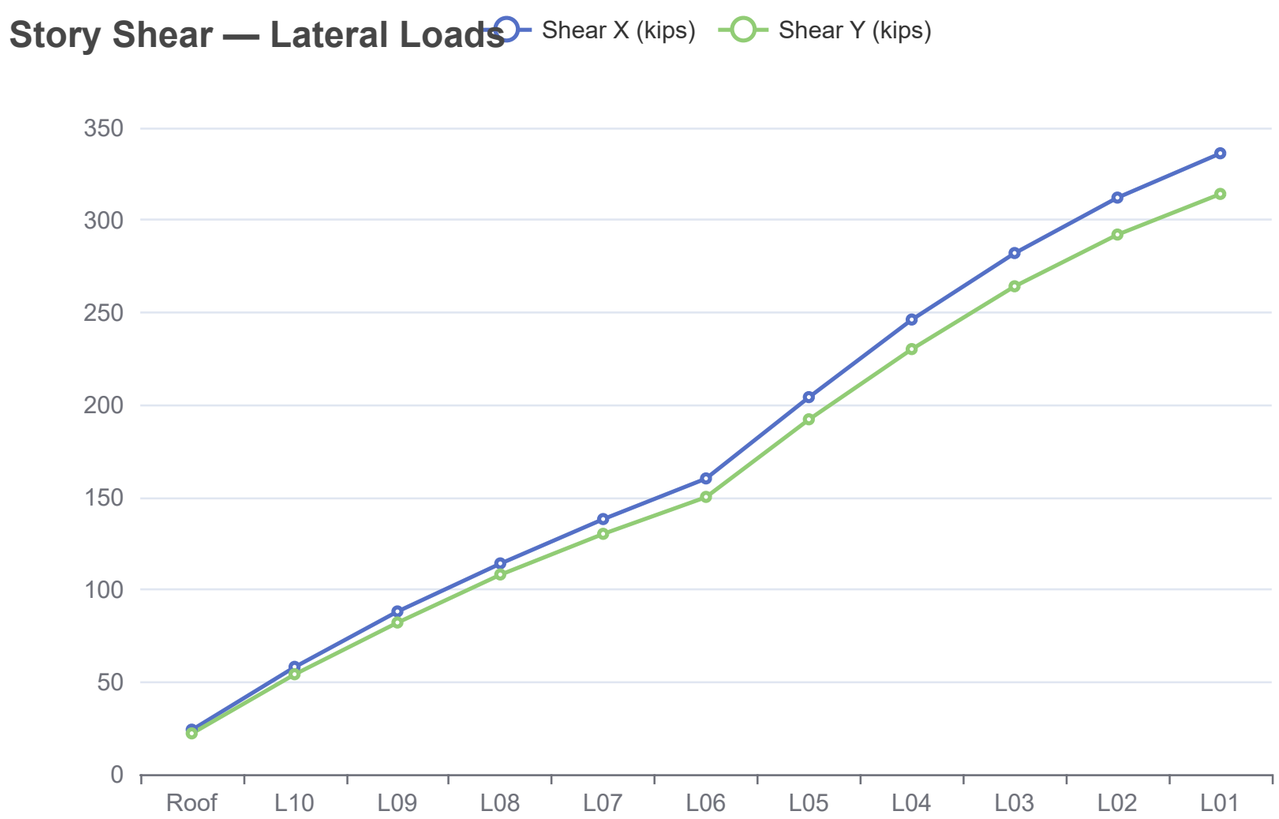
LATERAL LOAD ANALYSIS



Story Shear — X and Y Directions

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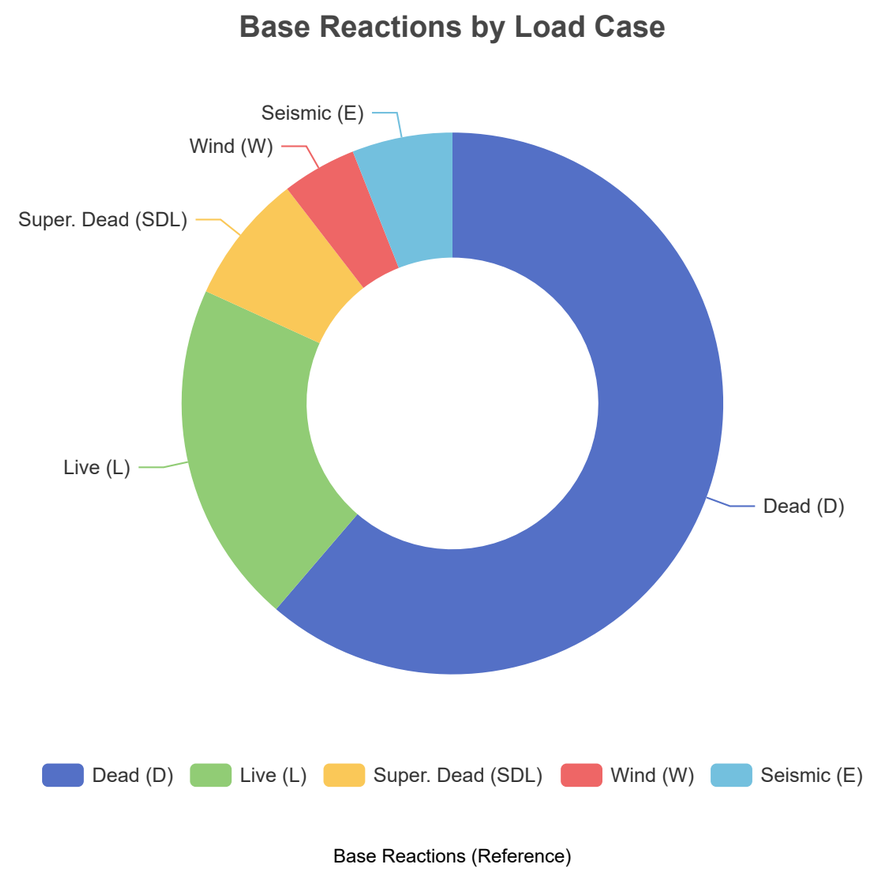
LOAD SUMMARY

Load Summary

Load Case	Fx (kips)	Fy (kips)	Fz (kips)
Dead (D)	0.0	0.0	2450.0
Live (L)	0.0	0.0	820.0
Super. DL (SDL)	0.0	0.0	310.0
Wind X (Wx)	185.0	0.0	12.0
Wind Y (Wy)	0.0	172.0	8.0
Seismic X (Ex)	240.0	0.0	18.0
Seismic Y (Ey)	0.0	228.0	15.0
1.2D+1.6L	0.0	0.0	4252.0
1.2D+1.0Ex+0.3Ey	72.0	228.0	2988.0



DESIGN CALCULATIONS

Dead Load — Slab Self Weight

1. Concrete unit weight

$$\gamma_c = 150 \text{ pcf}$$

Normal weight concrete per ACI 318-14 §19.2.3

2. Slab thickness

$$h_s = 8 \text{ in} = 0.667 \text{ ft}$$

1. Slab dead load

$$w_{DL} = \gamma_c \times h_s$$

→

$$w_{DL} = 150 \times 0.667 = 100 \text{ psf}$$

2. Superimposed dead load (MEP + finishes)

$$w_{SDL} = 25 \text{ psf}$$

Per architectural finish schedule

Live Load

1. Office occupancy (ASCE 7-22 Table 4.3-1)

$$L_o = 50 \text{ psf}$$

2. Tributary area

$$A_T = 28 \text{ ft} \times 30 \text{ ft} = 840 \text{ft}^2$$

1. Live load reduction (ASCE 7-22 §4.7)

$$L = L_o\left(0.25 + \frac{15}{\sqrt{K_{LL} A_T}}\right)$$

→

$$L = 50\left(0.25 + \frac{15}{\sqrt{2 \times 840}}\right) = 38.9 \text{ psf}$$

K_LL = 2 for two-way slabs

Factored Load (LRFD)

1. Governing combo: 1.2D + 1.6L (ASCE 7-22 §2.3.1)

$$w_u = 1.2 w_{DL} + 1.2 w_{SDL} + 1.6 L$$

→

$$w_u = 1.2(100) + 1.2(25) + 1.6(38.9) = 212.2 \text{ psf}$$

1. Factored column load

$$P_u = w_u \times A_T$$

→

$$P_u = 212.2 \times \frac{840}{1000} = 178.3 \text{ kips}$$

Lateral — Story Shear

1. Base shear (ASCE 7-22 §12.8)

$$V = C_s W$$

→

$$V = 0.094 \times 3580 = 336.5 \text{ kips}$$

Cs = SDS / (R/Ie) = 0.9 / (6/1.0) = 0.150 → 0.094 governs

1. Story force distribution (§12.8.3)

$$F_x = C_{vx} V, \quad C_{vx} = \frac{w_x h_x^k}{\sum w_i h_i^k}$$

k = 1.33 (interpolated, T = 0.66 s)

DEMAND-CAPACITY RATIOS

Element ID	Demand	Capacity	DCR	Status	Format
BEAM-001	372.5 k-ft	613.8 k-ft	**0.61**	OK	W18x50
SLAB-002	433.6 psf	815.2 psf	**0.53**	OK	8in
FTG-003	149.2 kips	152.9 kips	**0.98**	Check	Standard
FTG-004	203.2 kips	236.3 kips	**0.86**	OK	Standard
BEAM-005	545.9 k-ft	885.7 k-ft	**0.62**	OK	W18x50
BEAM-006	541.4 k-ft	846.3 k-ft	**0.64**	OK	W18x50
SLAB-007	62.4 psf	137.7 psf	**0.45**	OK	8in
COL-008	325.3 kips	524.5 kips	**0.62**	OK	HSS12x12
BEAM-009	440.1 k-ft	488.7 k-ft	**0.90**	OK	W18x50
SLAB-010	519.5 psf	920.4 psf	**0.56**	OK	8in
BEAM-011	796.9 k-ft	931.8 k-ft	**0.86**	OK	W18x50
FTG-012	260.9 kips	363.6 kips	**0.72**	OK	Standard
COL-013	683.6 kips	816.6 kips	**0.84**	OK	HSS12x12
FTG-014	334.9 kips	404.7 kips	**0.83**	OK	Standard
WALL-015	268.8 plf	449.3 plf	**0.60**	OK	12in CMU
COL-016	469.4 kips	793.0 kips	**0.59**	OK	HSS12x12
COL-017	97.8 kips	162.2 kips	**0.60**	OK	HSS12x12
FTG-018	694.1 kips	928.7 kips	**0.75**	OK	Standard
COL-019	221.9 kips	241.1 kips	**0.92**	OK	HSS12x12
WALL-020	630.5 plf	686.6 plf	**0.92**	OK	12in CMU
SLAB-021	132.4 psf	218.0 psf	**0.61**	OK	8in
COL-022	281.9 kips	452.1 kips	**0.62**	OK	HSS12x12

Color code: ≤ 0.85 (OK) 0.85–0.94 0.95–0.99 ≥ 1.0 (FAIL)

DEMAND-CAPACITY RATIOS (CONTINUED)

Element ID	Demand	Capacity	DCR	Status	Format
WALL-023	308.9 plf	568.1 plf	**0.54**	OK	12in CMU
COL-024	277.6 kips	488.3 kips	**0.57**	OK	HSS12x12
SLAB-025	175.1 psf	187.4 psf	**0.93**	OK	8in
WALL-026	523.8 plf	623.4 plf	**0.84**	OK	12in CMU
SLAB-027	446.7 psf	596.1 psf	**0.75**	OK	8in
FTG-028	365.5 kips	755.4 kips	**0.48**	OK	Standard
COL-029	217.2 kips	231.6 kips	**0.94**	OK	HSS12x12
COL-030	430.6 kips	543.4 kips	**0.79**	OK	HSS12x12